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A hybrid knowledge-based machine learning approach for ticket classification in the AWS cloud

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Abstract

Automated classification of customer support tickets is vital for modern service desks to reduce workload and improve response times. Machine learning (ML) methods provide flexibility and accuracy but often lack explainability and incur high computational costs. Knowledge-based systems (KBS) offer transparency and determinism but struggle with ambiguous or unseen inputs.

This paper presents a hybrid approach combining a lightweight rule-based classifier with a fallback ML model, deployed in a scalable serverless AWS environment. We evaluated logistic regression with TF-IDF, sentence embedding classifiers, and the hybrid model on a curated multilingual support dataset using accuracy, F1 score, precision, and recall.

Logistic regression outperformed embedding-based classifiers, achieving the highest macro-averaged F1 scores. The hybrid system leverages domain-specific rules to handle high-confidence cases and defers to ML when rules do not apply, consistently outperforming standalone ML models. Additional experiments tested robustness across multiple runs, train-test splits, and cross-lingual generalization with German data.

Deployed using AWS Lambda, S3, and DynamoDB, the solution balances low latency and scalability. The results demonstrate the hybrid approach's practical advantages for reliable, efficient support ticket classification.

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1. Introduction

In recent years, the increasing volume, linguistic diversity, and complexity of customer support tickets have made manual triage a significant bottleneck for service-oriented organizations. As businesses strive to maintain rapid response times and consistent quality of service, many have adopted machine learning (ML)-based solutions to automate the classification of incoming support requests. These models, particularly those built on deep learning architectures, often achieve high accuracy and flexibility, adapting to new patterns and data with minimal human intervention [1]. However, such approaches come with notable trade offs, they are computationally expensive, opaque in their decision-making, and may perform inconsistently in edge cases or underrepresented categories, especially in noisy or multilingual contexts [2].

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In contrast, knowledge-based systems (KBS) leverage curated rules and domain expertise to provide deterministic and interpretable outputs [3]. These systems require fewer computational resources and offer clear decision traceability, a quality particularly valuable in regulated industries or environments demanding high accountability. However, their rigidity and dependence on manually maintained rule sets make them poorly suited for handling ambiguous or informal language, which is common in real-world support communication.

Neither paradigm on its own fully addresses the challenges posed by dynamic, multilingual, and imbalanced support environments [4]. To bridge this gap, we propose a hybrid classification framework that combines the interpretability of rule-based systems with the adaptability of statistical models. In this setup, incoming tickets are first evaluated against a lightweight keyword-based rule engine [8, 9]. If a confident rule match is detected, a category is assigned immediately. Otherwise, the ticket is passed to a machine learning model specifically, a logistic regression classifier trained on either TF-IDF or sentence embeddings.

To ensure the proposed solution is scalable, cost-efficient, and easily maintainable, it is deployed within a fully serverless architecture on Amazon Web Services (AWS). Core components include Amazon S3 for data storage, AWS Glue for preprocessing, DynamoDB for rule management, and AWS Lambda for both model creation and inference. This architecture enables modular development, seamless integration, and on-demand execution with minimal operational overhead.

To validate the system, we performed a series of experiments across multiple language subsets (English and German), data sizes, and model configurations. IN addition, we analyze the impact of train-test splits and evaluate the system's generalization capabilities and robustness through repeated trials. These experiments were carefully designed to reflect real-world variability and edge cases, including scenarios with limited training data and class imbalance. By observing the consistency of results across different conditions, we are able to assess the stability and practical utility of the hybrid approach. The results support the hybrid model as a compelling and practical solution for real-world support automation, particularly in multilingual environments and under constrained computational settings.

The main scientific contribution of this paper is the evaluation of a lightweight hybrid classification framework that effectively integrates interpretable rule-based methods with statistical machine learning, deployed in a fully serverless cloud environment. We show that such integration improves classification performance in multilingual customer support scenarios, especially under data imbalance and limited resource constraints.

2. State of the Art

Automated classification of customer service tickets has attracted growing interest due to the increasing need for scalable and efficient triage systems. Prior research has extensively explored machine learning (ML) approaches to this problem, often relying on supervised learning with features such as TF-IDF, word embeddings, or neural language representations. Yaseen et al. [1] developed deep learning models leveraging recurrent architectures and stacked embeddings for natural language classification tasks, demonstrating high performance in specialized domains like medical text. However, the interpretability of such models remains limited.

The opacity of deep models is a significant concern, particularly in enterprise environments. Mani et al. [2] presented a comprehensive study on automating IT help desks, highlighting the challenges of deploying black-box models in production, especially in contexts that demand transparency and user trust.

In response to such limitations, researchers have begun to investigate methods that integrate structured knowledge or explicit rules into the decision process. Xu et al. [5] proposed combining knowledge graphs with retrieval-augmented generation (RAG) to improve question answering in customer service applications. Their approach enhances interpretability by linking outputs to grounded knowledge sources.

Rule-based systems, which offer deterministic and interpretable outputs, have a long standing history in AI. For example, Compton and Jansen [3] provided a foundational analysis of how expert systems acquire and refine rule sets based on domain expertise, a principle that continues to inform modern knowledge-based classifiers.

More recently, Krivosheev et al. [8] proposed "Active Hybrid Classification" a method that blends rule-based and statistical learning in an interactive framework, enabling adaptive decision making while preserving human-readable logic. Such hybrid systems offer a compelling trade-off between control and flexibility.

From an infrastructure perspective, Asogwa et al. [9] analyzed the use of hybrid machine learning techniques for text classification in big data contexts, emphasizing the importance of lightweight and scalable solutions. In parallel,

evaluation metrics such as accuracy, precision, recall, and F1-score have become standard for assessing classification models [12]. These are typically computed using libraries like scikit-learn [11], which provide consistent implementations for reproducible model comparison.

Building upon these developments, we propose a hybrid classification system that fuses interpretable rules with statistical models and deploys the entire pipeline in a modular, serverless architecture. This design emphasizes explainability, scalability, and multilingual support - key challenges in modern service automation.

3. Methodology

To evaluate the hybrid classification approach, we used a multilingual customer support dataset sourced from Kaggle. Our study focused on two language subsets, English and German, and narrowed the scope to three high-frequency support categories: Billing and Payments, Customer Service, and Technical Support. This ensured a consistent framework for comparing baseline models, hybrid configurations, and multilingual performance.

3.1. Data Preprocessing

We began by merging the subject and body fields of each support ticket into a unified text input, capturing both concise issue summaries and descriptive problem statements. The combined text was preprocessed using standard NLP techniques: lowercasing, removal of non-alphabetic characters, and elimination of stopwords using a multilingual stopword list compiled from NLTK resources [10]. This helped normalize syntax and reduce noise, especially for multilingual comparisons. Each language subset was filtered to retain only the target categories and stratified to preserve class balance across splits.

To assess the impact of training data volume, we performed additional experiments using multiple train-test splits: 80/20, 70/30, and 67/33. For each configuration, we repeated the evaluation five times with different random seeds to account for performance variability.

3.2. Baseline Models

We implemented two machine learning baselines to serve as reference points for evaluating the hybrid classifier:

- **TF-IDF + Logistic Regression**: Text was vectorized using a TF-IDF representation, considering both unigrams and bigrams with a cap of 1,000 features. A logistic regression classifier was trained on the resulting feature matrix.
- **Sentence Embeddings + Logistic Regression**: We used the `all-MiniLM-L6-v2` model from SentenceTransformers to encode ticket texts into dense semantic vectors. These embeddings were then used to train a logistic regression model with balanced class weights.

3.3. Rule-Based Classifier

To enhance interpretability and capture domain-specific signals, we developed a lightweight rule-based classifier. For each support category, the top TF-IDF-weighted keywords were extracted from training data. Manually curated rules were derived from these keywords to classify tickets if specific terms appeared in the text. This rule-based layer acted as a first filter in the hybrid system, returning a category prediction if a confident keyword match was found.

3.4. Hybrid Classification System

The hybrid classifier combined the rule-based and machine learning components. During inference, each ticket was first passed through the rule engine. If a rule triggered, the corresponding category was immediately assigned. If not, the ticket was processed by the ML model (TF-IDF + Logistic Regression or Sentence Embedding variant). This two-stage pipeline provided clarity in predictable cases and adaptability in more ambiguous ones.

3.5. Serverless System Architecture

To ensure scalability and cost-efficiency, the system was deployed using a fully serverless AWS architecture illustrated in Figure 1. Preprocessed data was stored in Amazon S3, rules were persisted in Amazon DynamoDB, and inference was handled via AWS Lambda. This architecture allowed modular updates to rule sets and models, ensured stateless execution, and reduced infrastructure overhead.

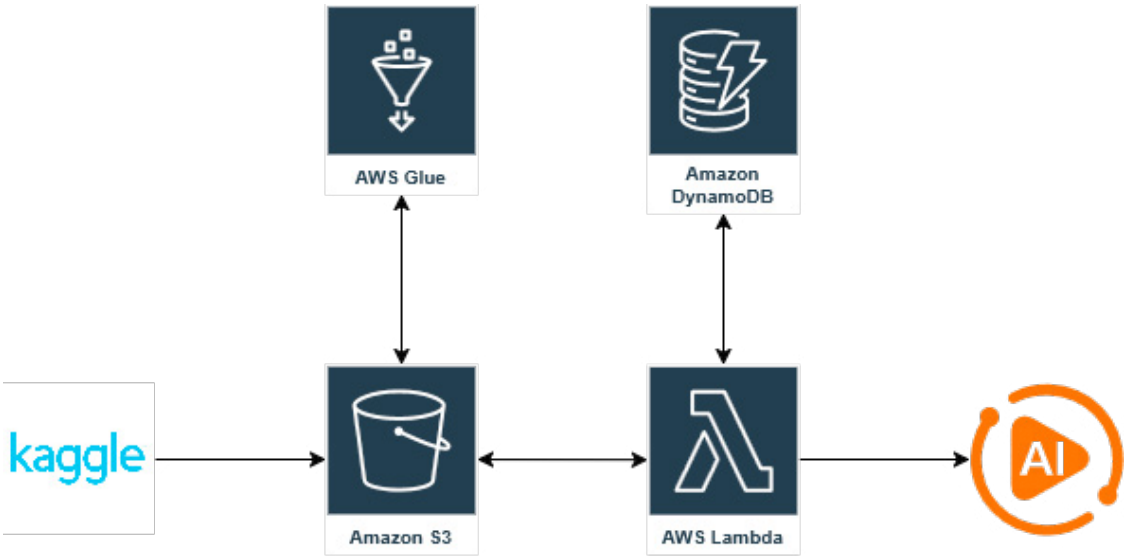

Fig. 1. AWS Serverless Architecture.

The serverless setup facilitated isolated evaluation of each classifier stage, supporting independent scaling and versioning of components.

3.6. Evaluation and Performance Analysis

Model performance was assessed across three dimensions:

1. **Language robustness**: comparing results on English and German subsets.
2. **Data scalability**: measuring performance across different train-test splits.
3. **Model stability**: repeating each experiment five times to compute average metrics and standard deviations.

The hybrid classifier was also compared against standalone models to evaluate gains in accuracy and interpretability. Additionally, we analyzed keyword importance and the distribution of rule-based versus ML-resolved cases to assess coverage and decision confidence.

All classification metrics (accuracy, precision, recall, F1-score) were calculated using the scikit-learn implementation [11], with macro and weighted averaging as appropriate [12].

4. Results

This section presents the evaluation results of the baseline models and the proposed hybrid classification system. We report both averaged metrics over repeated trials and detailed class-level scores to assess accuracy, robustness, and interpretability.

4.1. Experimental Setup

Experiments were implemented in Python using the `scikit-learn` and `sentence-transformers` libraries. The dataset included 11,923 English and 8,077 German customer support tickets. Each entry was labeled with one of three categories: Billing and Payments, Customer Service, or Technical Support.

For each language and model variant, we evaluated three different train-test splits: 80/20, 70/30, and 67/33. Every setup was repeated five times using different random seeds. Results presented here are averaged across runs to ensure stability and minimize randomness.

4.2. Average Metrics Summary

Table 1 presents the average performance metrics over five repeated runs with different seeds for each model, evaluated on the English subset using an 80/20 train-test split. The macro F1-score reflects the unweighted average of F1 scores across all classes, while the weighted F1-score takes class frequency into account.

Table 1. Averaged performance metrics.

Model	Accuracy	F1-macro	F1-weighted
TF-IDF + Logistic Regression	0.6976 ± 0.0087	0.6821 ± 0.0078	0.6928 ± 0.0085
Hybrid (Rules + TF-IDF)	**0.7176** ± 0.0072	**0.7115** ± 0.0104	**0.7156** ± 0.0076
Sentence Embeddings + Logistic Regression	0.6612 ± 0.0082	0.6626 ± 0.0056	0.6665 ± 0.0081

4.3. Per-Class Performance

Table 2 detailed class-level results revealed that the hybrid model particularly improved recall in underperforming categories such as Customer Service. For example, the rule-based component improved precision in the Billing and Payments, which had clearer lexical signals (e.g., “charge”, “billing”, “invoice”).

Table 2. Sample class-wise metrics (English, 80/20 split, Hybrid model).

Class	Precision	Recall	F1-score
Billing and Payments	0.94	0.67	0.78
Customer Service	0.56	0.48	0.53
Technical Support	0.71	0.84	0.77
Accuracy		0.71	

4.4. German Subset Evaluation

The classification models evaluated on the German language subset consistently achieved lower scores compared to their English language counterparts, which can be attributed to the smaller dataset size and increased lexical variability. Among the tested approaches, the hybrid classifier outperformed the two baselines in terms of both F1-macro and F1-weighted scores.

As shown in Table 3, the hybrid model achieved an accuracy of 0.6312 ± 0.0182 and a macro F1-score of 0.5995 ± 0.0183, outperforming the TF-IDF baseline and the sentence embedding model. While the absolute gains were modest, the hybrid approach demonstrated better balance across categories, particularly in scenarios with limited training data. This reinforces its potential for real-world multilingual deployments, especially in support environments where resources in languages other than English may be sparse.

Table 3. Averaged performance metrics over 5 runs (German, 80/20 split).

Model	Accuracy	F1-macro	F1-weighted
TF-IDF + Logistic Regression	0.6272 ± 0.0130	0.5665 ± 0.0171	0.6106 ± 0.0136
Hybrid (Rules + TF-IDF)	**0.6312** ± 0.0182	**0.5995** ± 0.0183	**0.6344** ± 0.0171
Sentence Embeddings + Logistic Regression	0.6049 ± 0.0108	0.5886 ± 0.0094	0.6113 ± 0.0107

4.5. Keyword-Based Rule Effectiveness

TF-IDF-based keyword extraction yielded interpretable term lists per category. These were manually refined to construct rule templates. Table 4 summarizes top extracted terms, many of which directly contributed to rule-based matches in the hybrid system.

Table 4. Top 5 TF-IDF keywords for each category.

Category	Top Keywords
Billing and Payments	billing, charge, payment, bill, refund
Customer Service	regulations, compliance, partnership, policy, update
Technical Support	server, connection, error, device, reboot

4.6. Impact of Train-Test Split

As illustrated in Figure 2 results from 70/30 and 67/33 train-test splits showed slight but consistent reductions in F1 scores, reflecting the expected impact of the decreased availability of training data. However, the hybrid model demonstrated a higher degree of resilience to these changes. The rule-based component served as a stabilizing mechanism, especially in categories with well-defined lexical features.

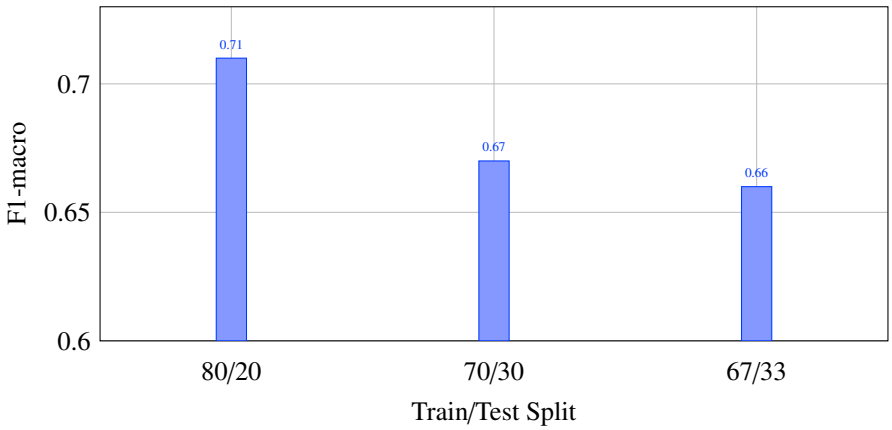

Fig. 2. F1-score of hybrid model across different train/test splits.

For instance, performance in the Billing and Payments category remained largely unaffected by the split ratio, thanks to consistent rule matches. Meanwhile, models relying solely on statistical learning showed a more pronounced performance drop, particularly in underrepresented classes such as Customer Service. These findings suggest that hybridization can effectively mitigate the risks associated with data sparsity and improve model robustness in real-world scenarios where annotated training data is limited.

4.7. Multilingual Robustness

Despite inherent differences in vocabulary, syntax, and data size, the hybrid classification system maintained its relative advantage when applied to the German subset. Although all models performed slightly worse in German than in English, the hybrid approach consistently delivered superior macro F1-scores and maintained balanced performance across categories.

This cross-lingual stability highlights the value of incorporating language-agnostic features, such as rules based on common domain terminology and sentence embeddings derived from multilingual transformer models. In support environments where datasets may be unevenly distributed across regions or languages, the hybrid model provides a flexible and adaptable framework that can operate effectively even under resource constraints. As such, it presents a compelling option for organizations managing multilingual support operations.

5. Discussion

Our findings confirm that the proposed hybrid classification system achieves a practical balance between interpretability and predictive performance, offering a practical alternative to purely statistical or rule-based approaches. While the TF-IDF + Logistic Regression baseline performed well overall, the addition of rule-based logic improved consistency across class distributions and led to a noticeable increase in macro-averaged F1 scores.

One of the primary advantages of the hybrid model lies in its ability to handle well-defined, lexically distinct queries, such as those in the Billing and Payments category, through direct rule matches. These cases often benefit from deterministic logic, reducing reliance on the statistical model and ensuring transparent decisions. This was particularly evident in the precision increase observed in the hybrid configuration, which leveraged high-confidence keyword triggers.

Conversely, in more abstract categories like Customer Service, which often involve nuanced language and broader semantics, the effectiveness of keyword-based rules diminished. Here, transformer-based sentence embeddings showed relative strength, suggesting that deep semantic representations are better suited to capturing contextual variability. However, this came at the cost of lower overall accuracy and increased computational complexity.

The hybrid system also proved more resilient to variations in training data volume. Across alternative train-test splits (70/30 and 67/33), its performance degradation was minimal, owing to the stability provided by the rule layer. Additionally, evaluations on the German subset affirmed the system's cross-lingual generalization capability, despite differences in vocabulary and label distribution.

From a deployment perspective, the modular architecture enabled by AWS Lambda and DynamoDB allows seamless updates to the rule set without retraining the ML model. This decoupling is especially valuable in production environments where ticket patterns evolve frequently, and model updates may be delayed.

Nonetheless, the rule engine remains a simplification of true linguistic understanding. Rules are manually crafted, lack syntactic structure, and do not account for negations or complex dependencies. Future improvements could involve automated rule induction, dependency parsing, or integration with knowledge graphs to enhance reasoning capabilities [6].

In summary, the hybrid model delivers interpretable, adaptive, and language-agnostic ticket classification. It is especially well-suited to environments where transparency and operational simplicity are valued alongside competitive performance.

6. Conclusion

This study evaluated a hybrid classification architecture to train customer support tickets in multilingual and imbalanced datasets. By combining lightweight, domain-specific rules with traditional machine learning techniques, logistic regression using TF-IDF and sentence embeddings, we aimed to strike a balance between predictive performance and interpretability.

The experiments demonstrate the measurable advantages of hybrid models compared to standalone ML approaches. The rule layer enabled high-confidence predictions in lexically consistent categories such as Billing and Payments, while the fallback to ML ensured adaptability for more ambiguous queries. Across both English and German subsets,

the hybrid system consistently outperformed individual baselines in accuracy, macro-averaged F1, and robustness to data variation.

From an engineering standpoint, deployment in a fully serverless AWS architecture proved highly effective. Modular separation of rules and models supported operational flexibility and cost efficiency. The system's ability to generalize across languages and tolerate shifts in training data size highlights its real-world applicability, especially for support teams operating in multilingual environments with limited resources.

However, challenges remain. Rule development was manual and did not capture syntactic or contextual nuances. As a result, rule coverage was incomplete in abstract domains such as Customer Service. To overcome these limitations, future iterations should explore dynamic rule induction, knowledge graph integration, and context-aware processing using more advanced NLP techniques.

Future work will expand on these findings by incorporating more support categories, testing on other public and proprietary datasets, and refining the hybrid logic through human-in-the-loop feedback. Ultimately, this research lays the groundwork for interpretable, scalable, and linguistically adaptive support automation systems.

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